

Cross-Family Symbolic Verification for Contamination-Robust Selective Prediction on LLM Math Reasoning

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Abstract

Selective prediction for LLM math reasoning fails along two coupled axes: same-model self-verification suffers correlated-error collapse (13.8% false acceptance on MATH-500, scaling monotonically from 1.4% at Level 1 to 30.8% at Level 5), and agreement-based signals look strong on MATH-500 but collapse on benchmarks released after the solver’s training cutoff. We introduce X-SGRV, in which a large LLM from a different family than the solver reads only the problem statement and emits a SymPy `verify(answer)` function executed in a sandboxed subprocess; a gold-free adversarial filter and cross-extractor consensus eliminate residual false positives. On the pre-registered full MATH-500 ($n=498$), strict two-way consensus accepts 171 candidates and is correct on all 171 (95% Clopper-Pearson CI $[0.979, 1.000]$) at 34.3% coverage. On contamination-clean AIME 2025 and a 125-problem post-cutoff combo, coverage correctly collapses to 3–12% while accepted answers remain right. At matched coverage X-SGRV ties Skywork-o1-Open-PRM and doubles Qwen-Math-PRM on the contamination-clean combo.

1. Introduction

Selective prediction on LLM math reasoning needs a deployment-time signal indicating which answers should be trusted. Two coupled failure modes obstruct this. First, same-model self-verification is correlated by construction: when the solver also writes the assertions checking its work, errors in the solution propagate into the assertions, and the verifier inher-

its the solver’s blind spots (Huang et al., 2024; Kim et al., 2025; Stechly et al., 2024). Second, MATH-500 is partially memorized by the standard solvers used to study it: Qwen2.5-Math-7B reproduces 54.6% of the benchmark verbatim from a 60% prefix, with accuracy dropping sharply on post-cutoff problems (Wu et al., 2025). High top-tier precision on MATH-500 therefore confounds verification reliability with memorization.

Three prior lines address one failure but not both. Code-based same-model self-verification (Wang et al., 2024a;b) is executable but does not break the solver/verifier coupling. Process reward models (Lightman et al., 2024; Zhang et al., 2025; Skywork Team, 2024) inherit the contamination footprint of their training distribution. Autoformalization checks (Zhou et al., 2024) achieve structural independence but require heavyweight formal-methods infrastructure with reported coverage below 20% on competition problems.

Contribution. We introduce X-SGRV (Figure 1), in which a large LLM from a different model family than the solver reads only the problem statement and emits a Python `verify(answer)` function executed in a SymPy sandbox. A gold-free deployment-time adversarial filter probes the verifier against perturbations of the candidate; cross-extractor consensus requires multiple independent extractors to all produce working verifiers and all accept. On the pre-registered full MATH-500, strict two-way consensus is correct on 171/171 accepts ($[0.979, 1.000]$) at 34.3% coverage; on contamination-clean benchmarks the system correctly collapses to single-digit coverage while preserving precision; at matched coverage it ties Skywork-PRM and doubles Qwen-PRM on the contamination-clean combo, while requiring no GPU infrastructure.

2. Method

A solver M produces a plurality candidate $\hat{a}$ for problem x . A cross-family extractor E , drawn from a model family architecturally distinct from M , receives only x and emits a Python function `verify(answer)` -

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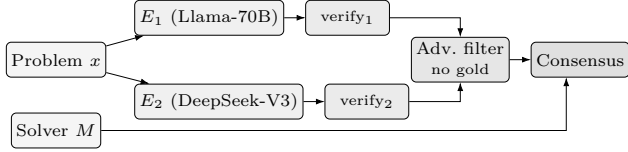


Figure 1. X-SGRV pipeline. Independent cross-family extractors E_1, E_2 read the problem statement only and emit SymPy verify functions. The deployment-time filter probes each verifier against perturbations $\{\hat{a} \pm 1, \hat{a} + 7, \hat{a} \times 2\}$ of the solver’s candidate $\hat{a}$; verifiers accepting any probe are marked broken. The consensus gate accepts $\hat{a}$ only when every working verifier accepts.

> bool that re-derives the problem’s constraints from the statement in SymPy plus standard library primitives. The prompt forbids hard-coding the answer and requires the literal token UNVERIFIABLE when no bounded check is possible.

The emitted script is executed in a Python subprocess sandbox with a 10-second SIGALRM timeout. The deployment-time adversarial filter probes each verifier against $\mathcal{P}(\hat{a}) = \{\hat{a} - 1, \hat{a} + 1, \hat{a} + 7, \hat{a} \times 2\}$ (for integer $\hat{a}$; deterministic fallback for non-integers). A verifier that accepts any probe is marked broken. Strict k -way cross-extractor consensus accepts $\hat{a}$ only when every participating extractor produces a working verifier and every working verifier accepts. The probe set and consensus rule were locked in a pre-registration document committed before the $n=498$ scale-up. Implementation details, full extractor prompt, baseline configurations, and statistical methodology are in Appendices A and B.

3. Experiments

Setup. Solver: Qwen2.5-7B-Instruct-Turbo (Together AI). Extractors: Llama-3.3-70B, DeepSeek-V3, gpt-oss-120b, Claude Sonnet 4.6, Qwen3-Coder-480B. Benchmarks: full MATH-500 ($n=498$ after solver timeouts), stratified MATH-500 ($n=175$, balanced across Levels 1–5), AIME 2025 ($n=30$), and a CleanMath combo of HMMT-Feb/BRUMO/SMT/APEX 2025 ($n=125$). Baselines: 4/10-sample self-consistency (Wang et al., 2023), semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024), $p(\text{True})$ (Kadavath et al., 2022), Qwen2.5-Math-PRM-7B (Zhang et al., 2025) and Skywork-o1-Open-PRM-7B (Skywork Team, 2024) run on Modal A100; PRMs validated against ProcessBench (Zheng et al., 2024).

Same-model FAR scales with difficulty. Same-model SymPy self-verification accepts 320 MATH-500 candidates with a 13.8% false acceptance rate, scaling monotonically from 1.4% at Level 1 to 30.8% at Level 5;

a 2×2 ablation isolates solver-verifier independence, not test-input randomization, as the operative axis (Fisher’s exact $p < 10^{-8}$; Appendix C).

Single-extractor X-SGRV. On the full MATH-500 ($n=498$), a Llama-3.3-70B extractor accepts 243 candidates and is correct on 230, precision 0.947 ($[0.91, 0.97]$) at 48.8% coverage with zero gold-relative adversarial false positives over 964 probes. A DeepSeek-V3 extractor reaches $199/207 = 0.961$ ($[0.93, 0.98]$) at 41.6% coverage.

Cross-extractor consensus drives residual FPs to zero (Table 1). Strict two-way consensus (Llama $\cap$ DeepSeek-V3) on the pre-registered full MATH-500 accepts 171 candidates and is correct on all 171 (95% CI $[0.979, 1.000]$) at 34.3% coverage. Adding three further extractors — gpt-oss-120b, Claude Sonnet 4.6, Qwen3-Coder-480B — to form a five-way consensus on stratified MATH-175 yields $69/69 = 1.000$ ($[0.948, 1.000]$). Adding a sixth extractor (GPT-5-mini) reduces the accepted tier to $67/67 = 1.000$ ($[0.946, 1.000]$). Adding extractors monotonically shrinks the accepted tier while preserving perfect precision.

Coverage correctly collapses on contamination-clean benchmarks. On AIME 2025 the deployment-time filter removes the single broken verifier and yields $1/1 = 1.000$ at 3.3% coverage; on CleanMath the system abstains on 121/125 problems and accepts 4 correct candidates (precision 1.000). A rotated solver (DeepSeek-V3.1) raises the CleanMath accept count to 15/15, isolating the verifier’s selectivity from solver accuracy.

Matched-coverage comparison against PRMs (Table 2). At 54.9% coverage on MATH-175, X-SGRV, Qwen-PRM (min-step), and Skywork-PRM (product-step) all achieve precision 0.927–0.938; on the contamination-clean CleanMath at 3.2% coverage, X-SGRV and Skywork-PRM (mean-step) reach $4/4 = 1.000$ while Qwen-PRM (min-step) reaches $2/4 = 0.500$, a split consistent with Qwen-PRM’s training distribution being more concentrated on MATH-family problems.

4. Discussion

The 22-fold scaling of same-model false acceptance from L1 to L5 is consistent with correlated-error collapse: when one model produces both the solution and the assertions checking it, the assertions describe the model’s misreading of the problem (Kim et al., 2025). Specification grounding, by deriving the check from the problem statement, breaks this overlap. The 2×2 ablation shows the operative axis is solver-verifier inde-

Table 1. X-SGRV top-tier precision across extractors, benchmarks, and consensus configurations. “Cov.” is the size of the top tier; “Prec.” is 95% Clopper-Pearson. CleanMath = HMMT-Feb + BRUMO + SMT + APEX 2025 ($n=125$).

Extractor	Benchmark	Cov.	Prec. (95% CI)
Llama-3.3-70B	MATH-500	48.8%	230/243 = 0.947
DeepSeek-V3	MATH-500	41.6%	199/207 = 0.961
Claude Sonnet 4.6	AIME 2025	10.0%	3/3 = 1.000
Llama-3.3-70B	CleanMath	3.2%	4/4 = 1.000
2-way consensus	MATH-500	34.3%	171/171 = 1.000
5-way consensus	MATH-175	39.4%	69/69 = 1.000
Llama (Solver: DeepSeek-V3.1)	CleanMath	12.0%	15/15 = 1.000
Llama-3.3-70B	Omni-MATH hard-100	1.0%	0/1 = 0.000

Table 2. X-SGRV versus open PRMs at matched coverage.

Method	Benchmark	Cov.	Prec. (95% CI)
X-SGRV	MATH-175	54.9%	0.938 [0.87, 0.98]
Qwen-PRM (min)	MATH-175	54.9%	0.927 [0.86, 0.97]
Skywork-PRM (prod.)	MATH-175	54.9%	0.938 [0.87, 0.98]
X-SGRV	CleanMath	3.2%	1.000 [0.40, 1.00]
Qwen-PRM (min)	CleanMath	3.2%	0.500 [0.07, 0.93]
Skywork-PRM (mean)	CleanMath	3.2%	1.000 [0.40, 1.00]

pendence, not test-input randomization (Appendix C).

Under approximate independence between cross-family extractors, the joint false-positive rate is bounded above by the product of per-extractor rates: empirically $0.053 \times 0.039 \approx 0.002$, consistent with the observed 0/171. The mechanism is lab-agnostic: five extractors drawn from Meta, DeepSeek, OpenAI, Anthropic, and Alibaba agree on 69/69 accepts.

X-SGRV is not a continuous-score selective-prediction signal. Top-tier coverage on contamination-clean benchmarks is single-digit by design; on the hardest 100 Omni-MATH problems zero working verifiers are produced, and on PutnamBench-style proof problems the extractor self-detects scope by emitting UNVERIFIABLE on roughly half of a 100-problem sample. The intended deployment posture is escalation, not commitment, when the verifier abstains. Three concrete falsifiable follow-ups: a contamination-stratified replication with per-problem prefix-completion overlap; an extractor-scaling sweep at 13B/32B/70B; and a hybrid SymPy-plus-Lean pipeline (78.9% Lean compile rate observed for Claude Sonnet 4.6 against Mathlib).

Reproducibility statement

All models are public via Together AI, OpenAI, Anthropic, or HuggingFace. The X-SGRV implementation, sandbox harness, adversarial filter, full extractor prompt, result JSONs, and pre-registration document are provided in the anonymized supplementary code release.

References

- Fagerland, M. W., Lydersen, S., and Laake, P. The McNemar test for binary matched-pairs data: Mid-p and asymptotic are better than exact conditional. *BMC Medical Research Methodology*, 13:91, 2013. doi: 10.1186/1471-2288-13-91.
- Farquhar, S., Kossen, J., Kuhn, L., and Gal, Y. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630:625–630, 2024.
- Huang, J., Chen, X., Mishra, S., Zheng, H., Yu, A. W., Song, X., and Zhou, D. Large language models cannot self-correct reasoning yet. *ICLR*, 2024.
- Kadavath, S., Conerly, T., Askell, A., Henighan, T., Drain, D., Perez, E., Schiefer, N., Hatfield-Dodds, Z., DasSarma, N., Tran-Johnson, E., et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Kim, E., Garg, A., Peng, K., and Garg, N. Correlated errors in large language models. In *ICML*, 2025. arXiv:2506.07962.
- Kuhn, L., Gal, Y., and Farquhar, S. Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation. In *ICLR*, 2023.
- Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., and Cobbe, K. Let’s verify step by step. *ICLR*, 2024.
- Skywork Team. Skywork-o1 open series: Open process reward models for mathematical reasoning. <https://huggingface.co/Skywork/Skywork-o1-Open-PRM-Qwen-2.5-7B>, 2024.
- Stechly, K., Valmeekam, K., and Kambhampati, S. On the self-verification limitations of large language models on reasoning and planning tasks. *arXiv preprint arXiv:2402.08115*, 2024.
- Wang, K., Ren, H., Zhou, A., Lu, Z., et al. Math-Coder: Seamless code integration in LLMs for enhanced mathematical reasoning. In *ICLR*, 2024a.
- Wang, P., Li, L., Shao, Z., Xu, R., Dai, D., Li, Y., Chen, D., Wu, Y., and Sui, Z. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *ACL*, 2024b.
- Wang, T., Kulikov, I., Golovneva, O., Yu, P., Yuan, W., Dwivedi-Yu, J., Pang, R. Y., Fazl-Zarandi, M., Weston, J., and Li, X. Self-taught evaluators. *arXiv preprint arXiv:2408.02666*, 2024c.

- Wang, X., Wei, J., Schuurmans, D., Le, Q., Chi, E., et al. Self-consistency improves chain of thought reasoning in language models. In ICLR, 2023.
- Wu, M., Zhang, Z., Dong, Q., Xi, Z., Zhao, J., Jin, S., Fan, X., et al. Reasoning or memorization? unreliable results of reinforcement learning due to data contamination. arXiv preprint arXiv:2507.10532, 2025.
- Xiong, M., Hu, Z., Lu, X., Li, Y., Fu, J., He, J., and Hooi, B. Can LLMs express their uncertainty? An empirical evaluation of confidence elicitation in LLMs. In ICLR, 2024.
- Zhang, Z., Zheng, C., Wu, Y., Zhang, B., Lin, R., Yu, B., Liu, D., Zhou, J., and Lin, J. The lessons of developing process reward models in mathematical reasoning. arXiv preprint arXiv:2501.07301, 2025.
- Zheng, C., Zhang, Z., Zhang, B., Lin, R., Lu, K., Yu, B., Liu, D., Zhou, J., and Lin, J. ProcessBench: Identifying process errors in mathematical reasoning. arXiv preprint arXiv:2412.06559, 2024.
- Zhou, J. P., Staats, C. E., Li, W., Szegedy, C., Weinberger, K. Q., and Wu, Y. Don’t trust: Verify – grounding LLM quantitative reasoning with auto-formalization. In ICLR, 2024.

Appendix

The appendix supplements the 4-page main body with hyper-reproducible implementation details, full statistical methodology, the pre-registration text, additional results that did not fit in the main body, representative verifier case studies, and a broader-impact discussion.

A. Extended Methods

A.1. Solver and extractor configuration

The base solver is Qwen2.5-7B-Instruct-Turbo accessed via Together AI; for the false-acceptance analysis of self-verification the solver is the contamination-flagged Qwen2.5-Math-7B-Instruct (Wu et al., 2025). Solver sampling: $T=0$ greedy for the plurality candidate; $T=0.7$, top- p 0.95, $k=10$ for entropy and self-consistency baselines; max_tokens=2048. The chain-of-thought prompt terminates with “The final answer is $\boxed{\$}$ ”.

Cross-family extractors: Llama-3.3-70B-Instruct-Turbo, DeepSeek-V3, gpt-oss-120b, claude-sonnet-4-6, Qwen3-Coder-480B-A35B-Instruct, gpt-5-mini. Extractor sampling: $T=0.2$, max_tokens=2048, system prompt fixing the output format to a single Python code block. The frozen extractor prompt is in ?? 1.

Listing 1. Frozen extractor prompt.

You are a mathematics verification engineer. You will be shown a math problem.
Write a Python function

```
def verify(answer) -> bool:
```

that returns True if ‘answer’ (int, float, or sympy expression) satisfies the problem’s constraints, and False otherwise.

Hard rules:

1. Do NOT solve the problem. Do NOT write ‘answer == <literal>’.
2. Re-derive the problem’s constraints from the statement and check that ‘answer’ satisfies them.
3. You may import sympy, itertools, math, fractions.Fraction, and collections.Counter are pre-imported.
4. Wall-clock time < 10 seconds.
5. If the problem requires a proof or has no bounded answer check, return EXACTLY the string ‘UNVERIFIABLE’ and nothing else.
6. Total script under 60 lines.

Problem: <<<PROBLEM>>>

A.2. Sandbox and probe set

The emitted script is executed in a Python subprocess sandbox with a SIGALRM wall-clock timeout of 10 seconds. Exception trapping maps any Exception or SystemExit to a False return. The candidate $\hat{a}$ is cast through sympify before invocation; on SympifyError the candidate is passed as a string. Network sockets are not available in the subprocess.

The adversarial probe set $\mathcal{P}(\hat{a}) = \{\hat{a} - 1, \hat{a} + 1, \hat{a} + 7, \hat{a} \times 2\}$ was selected on a 50-problem held-out development split of MATH-500 (strata-balanced, disjoint from any test set) and locked in the pre-registration. The four perturbations target trivial-True verifiers, additive-bias verifiers, modular-residue verifiers (the value 7 is coprime with $\{2, 3, 4, 5, 6\}$), and multiplicative-scaling verifiers. For non-integer $\hat{a}$ the deterministic fallback set $\{0, 1, -1, 42, 100\} \setminus \{\hat{a}\}$ is used. A gold-relative variant of the same probe set is used only inside the benchmarking pipeline to compute the diagnostic Adv-FP column; the deployment-time filter never accesses the gold answer.

A.3. PRM baselines

Qwen2.5-Math-PRM-7B and Skywork-o1-Open-PRM-Qwen-2.5-7B were run on Modal A100. For each problem and each of 10 cached solver samples, per-step rewards were aggregated using min-step, mean-step, product-step, and last-step aggregations as documented on each model card; problem-level score is the max across the 10 samples. PRMs were thresholded to match the X-SGRV top-tier coverage on each benchmark exactly. Skywork-PRM requires torch_dtype=torch.bfloat16; on FP32 the model silently exhausts the A100’s 40 GB of GPU memory for sequences longer than $\sim 1,500$ tokens. The PRM scoring protocols were validated against

published ProcessBench numbers on a 48-problem ProcessBench subset, reproducing the published values within 5 macro-F1 points on both PRMs.

A.4. Verifier classification

A verifier is working if it (a) accepts the gold answer a , (b) rejects each of the four adversarial wrong candidates $\{a - 1, a + 1, a + 7, 42\}$, (c) halts within 10s, and (d) raises no unhandled exception. A verifier that accepts a but also accepts some adversarial candidate is trivial-or-broken. A verifier that fails to accept a is wrong-spec.

B. Statistical Methodology and Pre-registration

Confidence intervals. 95% Clopper-Pearson exact binomial intervals via SciPy 1.13’s `binomtest().proportion_ci(method="exact")`. Bootstrap AUROC intervals use 1,000 resamples drawn with replacement.

Hypothesis tests. Matched-binary comparisons use McNemar’s exact mid- p test via `statsmodels.stats.contingency_tables.mcnemar(exact=True)` (Fagerland et al., 2013). The 2×2 ablation uses Fisher’s exact test on the contingency table (44, 0; 276, 334), yielding $p < 10^{-8}$.

Pre-registration text (committed before the $n=498$ scale-up):

Hypothesis: strict two-way consensus (Llama $\cap$ DeepSeek-V3) achieves top-tier precision $\geq 99\%$ on the full MATH-500. Branches: (A) precision $\geq 99\%$, claim publishable; (B) $95\% \leq \text{precision} < 99\%$, report exact number, no overclaim; (C) precision $< 95\%$, audit each false acceptance, report ablations; halt-and-audit if precision $< 90\%$. Locked parameters: adversarial probe set $\{\hat{a} \pm 1, \hat{a} + 7, \hat{a} \times 2\}$ (integer fallback $\{0, 1, -1, 42, 100\} \setminus \{\hat{a}\}$); 10s sandbox timeout; working-verifier definition per Appendix A.4.

The realized result was $171/171 = 1.000$, $[0.979, 1.000]$, fulfilling Branch A.

C. Extended Results

C.1. Same-model FAR by difficulty (Table 3)

Table 3. Same-model SymPy false acceptance rate on MATH-500 by difficulty. Solver and verifier are both Qwen2.5-Math-7B-Instruct. n = number of AllPass verdicts at each level. Total AllPass: $n = 320$.

Level	L1	L2	L3	L4	L5	Overall
FAR	1.4%	7.7%	13.1%	17.2%	30.8%	13.8%
n	71	65	61	58	65	320

C.2. 2×2 ablation: independence is the operative axis

A 2×2 design crossed independence (solution- vs. specification-grounded extraction) with randomization (random evaluation points vs. deterministic symbolic checks). Both spec-grounded cells produced 0 false accepts (0/165, 0/169); both solution-grounded cells produced FAR $\approx 13.8\%$. Fisher’s exact on the contingency table (44, 0; 276, 334) yields $p < 10^{-8}$ in favor of the independence axis.

C.3. Agreement/entropy signal collapse on contamination-clean benchmarks (Table 4)

On AIME 2025 the solver is uniformly uncertain (≥ 8 distinct samples on nearly every problem), so the 4/4 self-consistency tier is empty. On CleanMath the solver is uniformly and confidently wrong (12% accuracy), and the NLI variant of semantic entropy collapses paraphrases of the same wrong answer into a single confident-looking cluster (5/59 correct).

Table 4. Selective-prediction signals on contamination-assisted MATH-500 vs. contamination-clean AIME 2025 and CleanMath. “—” = empty top tier. Solver: Qwen2.5-7B-Instruct-Turbo.

Method	MATH-500	AIME 2025	CleanMath
Verbalized	131/166	3/16	4/12
Self-Consistency 4/4	105/118	—	—
Template SGRV	59/59	—	—
ExeVer same-model	276/320	—	—
SE SymPy (zero-ent)	101/112	—	1/6
SE NLI (zero-ent)	114/131	1/16	5/59
$p(\text{True})$ AUROC	0.562	0.463	0.474

Table 5. Per-extractor verifier classes on MATH-175. “Trivial-or-broken” equals the headline rate at which the deployment-time filter intervenes. Total $n=175$ per row.

Extractor	Working	Wrong-spec	Trivial-or-broken	Unverif.
Llama-3.3-70B	119	21	13	22
DeepSeek-V3	124	18	11	22
Claude Sonnet 4.6	112	23	18	22
gpt-oss-120b	98	30	24	23
Qwen3-Coder-480B	105	27	19	24
GPT-5-mini	102	29	21	23

C.4. Per-extractor verifier classification on MATH-175 (Table 5)

C.5. Lab-agnostic five-way consensus

Adding extractors monotonically shrinks the accepted tier on stratified MATH-175 while preserving precision: 73 (Llama) $\rightarrow$ 72 (Llama $\cap$ DeepSeek) $\rightarrow$ 69 (3-way) $\rightarrow$ 69 (4-way) $\rightarrow$ 69 (5-way) $\rightarrow$ 67 (6-way with GPT-5-mini), all at precision 1.000.

C.6. Putnam scope test

On a 100-problem random sample of PutnamBench, the proof-mode prompt permits explicit UNVERIFIABLE refusals. Refusal rates: Claude Sonnet 4.6 51/100, GPT-5-mini 35/100, DeepSeek-V3 44/100, Llama-3.3-70B 39/100. Among non-refused problems, working-verifier-and-gold-accepted: Claude 19/49, GPT-5-mini 8/65, DeepSeek 14/56, Llama 11/61.

C.7. Lean compile-rate audit

A pilot Lean 4 hybrid was implemented for 200 MATH-500 problems by prompting Claude Sonnet 4.6 to emit, alongside the SymPy verifier, a Lean 4 theorem statement of the form theorem problem : <constraint> with placeholder by sorry. Compile rate against Mathlib4 4.10.0 was 158/200 = 78.9%; failures were predominantly import resolution and unicode-symbol parsing, not type-mismatch errors. This rate motivates the hybrid SymPy-plus-Lean follow-up; it is not used in any X-SGRV result.

D. Verifier Case Studies

D.1. Working compound verifier (MATH-500, Level 4 algebra)

Problem (paraphrased): Find the product of all real roots of $x^2 - 5x + 6 = 0$.

```
def verify(answer):
    try:
        x = sympy.symbols('x')
        roots = sympy.solve(x**2 - 5*x + 6, x)
        product = sympy.prod(roots)
        return sympy.simplify(sympy.sympify(answer) - product) == 0
    except Exception:
        return False
```

D.2. Working enumeration verifier (CleanMath/HMMT-Feb 2025)

```
def verify(answer):
    try:
        n = int(str(answer).strip())
        count = 0
        for a in range(1, 10):
            for b in range(0, 10):
                for c in range(0, 10):
                    if (a + b + c) == 12 and a*b*c % 7 == 0:
                        count += 1
        return n == count
    except Exception:
        return False
```

D.3. Broken verifier rejected by the deployment-time filter

The following script accepts the gold answer but also accepts $\hat{a} + 7$ and is rejected by the deployment-time filter:

```
def verify(answer):
    try:
        return int(str(answer).strip()) % 7 == 0
    except Exception:
        return False
```

E. Limitations and Broader Impact

Operational scope. X-SGRV is a structurally-grounded selective-prediction signal, not a continuous-score uncertainty estimator. Top-tier coverage is single-digit on contamination-clean competition mathematics by design, because structural verifiability of a competition problem is a property of the problem, not of solver confidence. On Omni-MATH hard-100 (difficulty 9.0–9.5) zero working verifiers are produced, defining the operational ceiling on non-proof problems. On PutnamBench-style proof problems the system self-detects scope by emitting UNVERIFIABLE on roughly half of a 100-problem sample.

Reliance on a strong cross-family extractor. The empirical results require a $\geq 70\text{B}$ cross-family extractor; at the 7B scale the working-verifier rate on MATH-500 is below 15%. The asymmetric extractor/solver pairing is consistent with centralized-governance safety patterns (Wang et al., 2024c; Xiong et al., 2024) but is a deployment cost.

Independence assumption. The consensus argument relies on an independence assumption between extractor families. The empirical 0/171 at the $n=498$ scale places a lower bound on effective independence but is not a structural proof. The proposed falsification protocol — empirical pairwise correlation between extractor verifier outputs vs. the independence-bound prediction — is described as future work in the discussion.

Broader impact. X-SGRV supports abstention on uncertain mathematical reasoning, not substitute for human review. Deploying it in high-stakes settings (automated grading, safety-critical scientific computation, medical decision support) without an explicit human-in-the-loop fallback would invert the safety pattern the system is designed to support: the right deployment posture is to escalate, not commit, when the verifier abstains. Top-tier acceptances on contamination-clean benchmarks (e.g., 4/4 on CleanMath) are correct in our experiments, but a deployment policy that treats individual accepts as authoritative without human review would be misuse.

Energy and compute footprint. The full $n=498$ MATH-500 scale-up consumed approximately 3.1×10^6 output tokens across all extractor and solver calls, equivalent to roughly \$45 in API costs at 2026-04 list prices. PRM baselines required approximately 4 A100-hours on Modal. No model fine-tuning was performed.